A Consensus-Trained U-Net with a Learned Component Scorer

A Solution to the Solar Filament Segmentation Challenge 2026

Mukharbek Organokov
mukharbek.organokov@gmail.com

ABSTRACT

This report describes our solution to the Solar Filament Segmentation Challenge 2026, a Kaggle competition on automatic segmentation of solar filaments in GONG H-α observations. A single U-Net with a ResNet-34 encoder, run at 1536 px, is trained on the consensus of the annotators and averaged over four flipped views. Its connected components are candidates, and a gradient-boosted component scorer, trained on out-of-fold candidates, predicts each candidate's expected contribution to Panoptic Quality (PQ) and keeps those that pay. On all 707 training images, scored out of fold with a split by calendar month, the pipeline reaches PQ 0.422 (95% bootstrap interval 0.412–0.431), against 0.409 for the same network at 1024 px with a hand-tuned rule; one MAGFiLO annotator scored against another on the same images reaches 0.340. The public leaderboard gives 0.35. We list the variants that did not help and show that most of the remaining loss is disagreement between annotators.

1 INTRODUCTION

This report goes over the details of our submitted solution to the Solar Filament Segmentation Challenge 2026 [2]. The challenge, launched on July 10, 2026, is organized by the Earth-Space AI Research (ESAIR) Lab[1]. This challenge utilizes the MAGFiLO dataset for evaluation of the segmentation algorithms [1].

One property of the data shapes every decision: the annotators disagree strongly, and each annotator is scored separately against the same prediction. A prediction can therefore never match all annotators, and the best achievable PQ is far below 1. Our solution is (i) a simple, reproducible segmenter trained on the annotators' consensus; (ii) a learned component scorer that decides which instances to submit by their expected PQ contribution; and (iii) an error analysis, with the annotator-vs-annotator score as a reference, that explains why the usual improvements did not help. Only the competition's training images and annotations are used; no external filament masks, spines or chirality labels. Code, weights and a notebook that reproduces the submission are in the repository, and every experiment is logged in `docs/DESIGN.md`.

2 METHODOLOGY

2.1 Data and metric

The training set has 707 images (2048×2048 px) with 8,199 annotated filaments; 411 images have one annotator, 145 two and 151 three, giving 1,154 annotator-image entries. The test set has 180 images. Within each entry, a ground-truth (GT) and a predicted mask form a true positive (TP) when their IoU exceeds 0.5, and PQ [7] is $\sum_{TP} \mathrm{IoU}/(|TP| + 0.5|FP| + 0.5|FN|)$, pooled over all entries. It factors into segmentation quality SQ (the mean IoU of the TPs) and recognition quality RQ. Our implementation reproduces the organisers' self-evaluation notebook exactly. That code does not enforce one-to-one matching, so overlapping predictions of one filament would each count as a TP; all our submissions are disjoint by construction.

2.2 Input, target and network

Each image is area-downsampled from 2048 to 1536 px and normalised with fixed constants; there is no other pre-processing. The training target is the consensus, i.e. the fraction of annotators marking each pixel, computed at 2048 px and then downsampled. With binary cross-entropy (BCE) on this soft target the network learns the probability that an annotator marks a pixel, which is the quantity the multi-annotator metric rewards. The network is a U-Net [10] from `segmentation_models_pytorch` [6] with an ImageNet ResNet-34 encoder [5] adapted to one input channel (24.4 M parameters). Loss: BCE + soft Dice. AdamW, learning rate $5 \cdot 10^{-4}$ with one epoch of warm-up and cosine decay, weight decay 10^{-4}, batch 4, 40 epochs. Augmentation: flips, 90-degree rotations, affine (±30°, scale 0.9–1.1), brightness, contrast and gamma. The submitted model is trained on all 707 images (36 min on one NVIDIA A100, about 3 h on an Apple M3 Max).

2.3 From probabilities to instances

At inference the probability is averaged over four views (identity, horizontal, vertical and both flips) and upsampled bilinearly to 2048 px. The map is thresholded at 0.45. A morphological closing with a 31 px disk is used only to decide which fragments belong to one filament; the masks keep the thresholded pixels. The 8-connected components of at least 100 px are the candidates, disjoint by construction (Fig. 1).

2.4 Component scorer

Every false positive costs half a point in the PQ denominator, so which candidates are submitted matters as much as their shape. A kept candidate that matches k of the n entries of its image adds about $k \cdot \mathrm{IoU}$ to the numerator and $0.5n$ to the denominator either way, so it pays when its expected matched IoU per entry exceeds about PQ/2. We therefore regress, for each out-of-fold candidate, the target $y = \frac{1}{n} \sum_e \mathrm{IoU}_e \, [\mathrm{IoU}_e > 0.5]$ over the entries e of its image, with scikit-learn's `HistGradientBoostingRegressor` [8] (400 iterations, learning rate 0.05, 31 leaves, at least 40 samples per leaf, L2 1.0). It uses 20 features: area and log-area; mean, maximum, standard deviation, 10/50/90% quantiles of the probability and the fraction above 0.8; elongation, length, width and compactness; the brightness inside and its contrast with a 15 px ring around the component; distance from the disk centre, latitude and longitude on a

[1] www.esairlab.com/

Kaggle Competition, 2026, kaggle.com/competitions/filament-segmentation-2026
2026.

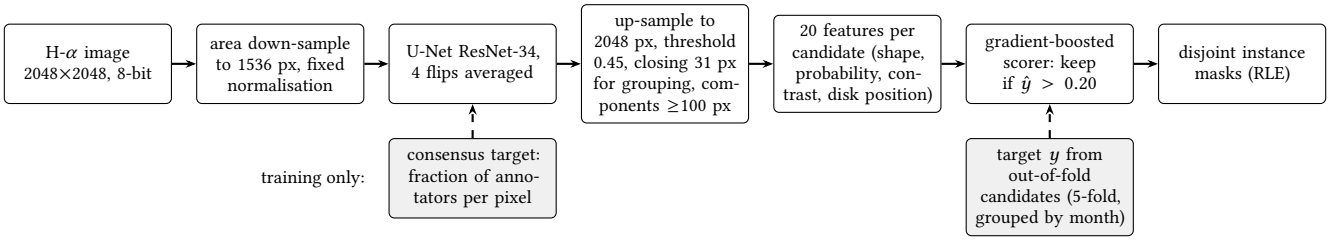


Figure 1: The end-to-end pipeline. White: inference, applied to every test image. Grey: what each learned stage is trained on.

disk fitted with a Hough circle; and two image-level features (number of candidates and mean probability). We keep candidates with $\hat{y} > \tau = 0.20$; PQ varies by less than 0.001 over $\tau = 0.15$–0.25. A hand rule (area ≥ 300 px and mean probability ≥ 0.8) was used before. The whole inference, features included, takes 4.5 s per image on an M3 Max.

2.5 Validation split and tuning

Filaments persist for days, so images from the same week are near-duplicates. We use a 5-fold `GroupKFold` grouped by calendar month (YYYYMM), so no month is in both training and validation. The threshold and closing are grid-searched with the exact metric on the pooled out-of-fold (OOF) probabilities of the five fold models. The scorer is cross-validated on the same folds: the scorer applied to a fold's candidates is trained on the other four, so every OOF number below comes from models that never saw the image. All choices were made on OOF, never on the public leaderboard, which covers about 90 test images and shows two decimals.

Table 1: Main results. OOF: out of fold, 707 images, 1,154 entries. Brackets: 95% bootstrap interval over images.

	PQ	SQ	RQ
Final pipeline, OOF	**0.422** [0.412, 0.431]	0.674	0.626
per fold, mean ± s.d.	0.422 ± 0.012		
296 multi-annotator images	0.423	0.671	0.630
Annotator vs. annotator (same 296)	0.340	0.634	0.536
1024 px, 2 views, hand rule, OOF	0.409 [0.399, 0.418]	0.666	0.614
+ 4 views	0.412	0.667	0.617
+ component scorer	0.417	0.668	0.624
1536 px, 4 views, hand rule	0.417	0.671	0.621
Gain over 1024 px rule (paired)	+0.013 [0.008, 0.018]		
Public leaderboard, final pipeline	0.35	–	–
Public leaderboard, 1024 px rule	0.35	–	–

3 EVALUATION

3.1 Main results

Table 1 gives the main numbers. Uncertainty comes from two sources: the spread of PQ over the five folds, and a bootstrap that resamples whole images (all entries of an image together, 2,000 resamples) and recomputes the pooled PQ. The paired bootstrap of the difference to the previous pipeline resamples the same images for both. To put the numbers in scale, each annotator's masks are scored as predictions against every other annotator of the same image (296 images with two or three annotators).

The final pipeline improves on the previous one on all five folds (per fold: +0.017, +0.007, +0.013, +0.012, +0.018), and the paired bootstrap interval of the gain excludes zero. Its components contribute unequally: tuned per fold, 1536 px beats 1024 px on four of five folds (+0.009, −0.002, +0.009, +0.005, +0.007), and 2048 px matches 1536 px on the two folds tried. The scorer beats the submitted hand rule (confidence ≥ 0.8) on every fold, by 0.002–0.009, but the best pooled hand rule (confidence ≥ 0.85, PQ 0.420) by only 0.002. The two leaderboard scores are equal at the two decimals shown: scaled to the 90 or so public images, the bootstrap interval of a single pipeline is about ±0.027, twice the gain. Local OOF ran 0.06–0.07 above the public leaderboard for every submission.

Fig. 3 shows the three evaluation plots requested by the organisers on the OOF predictions. Over GT/prediction pairs with IoU > 0, the mean IoU is 0.573 and the mean Dice 0.707 (0.553 and 0.688 for the 1024 px rule pipeline). 3.4% of predicted filaments overlap more than one annotation (merges) and 3.7% of annotations overlap more than one prediction (splits); 11.7% of predictions overlap no annotation of their entry and 23.3% of annotations overlap no prediction. Compared with the hand rule, the scorer removes a quarter of the false positives at the cost of more missed annotations, mostly filaments few annotators agree on. Fig. 2 shows typical images.

3.2 What did not help

Table 2 lists the variants we tried, each with its own tuned post-processing. A Mask R-CNN [4] detector learns grouping explicitly, but its masks are coarser (SQ 0.642 vs. 0.657 for a U-Net trained for a similar number of epochs); adding its extra instances, or a per-filament crop refiner at native resolution following Reche & Cid [9], changes PQ by less than 0.002 on the fully trained U-Net. A ResNet-50 encoder, an input divided by its own radial brightness profile (limb-darkening correction) and a Lovász hinge loss [3] match or underperform the baseline; with the corrected input the best threshold moves from 0.45 to 0.65, so the network becomes more confident but not more accurate. Averaging two equal-strength models with different encoders gives the PQ of the better one, and the 5-fold average gives the same leaderboard score as one model: the models share their errors because the errors come from the annotators, so the submission is one network.

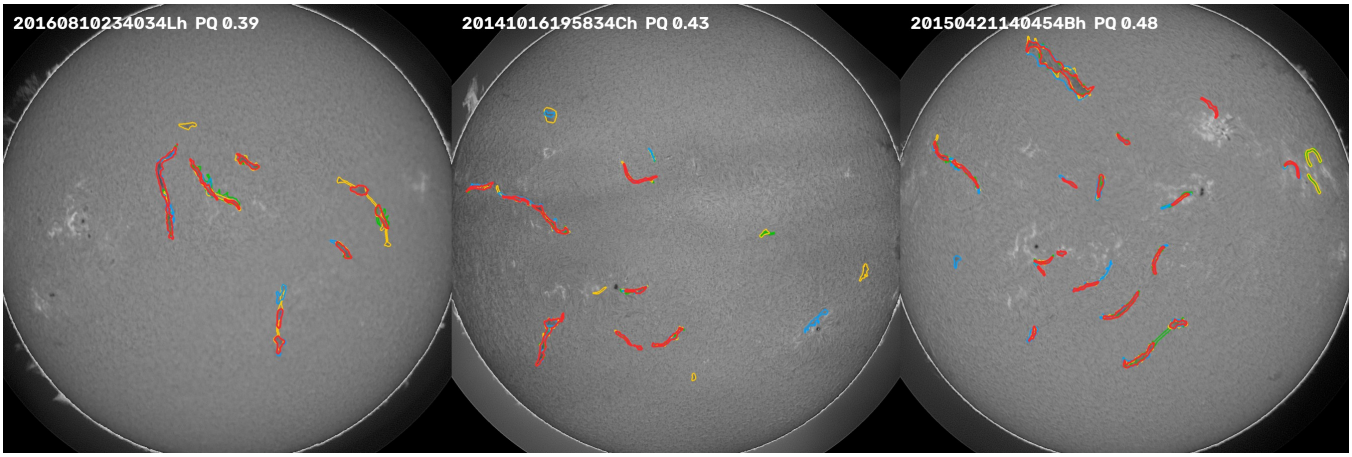


Figure 2: Out-of-fold predictions of the final pipeline (red) on three validation images with three annotators (green, yellow, blue), at the 50th, 75th and 90th percentile of image PQ among images with at least five filaments per annotator. Many misses are filaments marked by one annotator only.

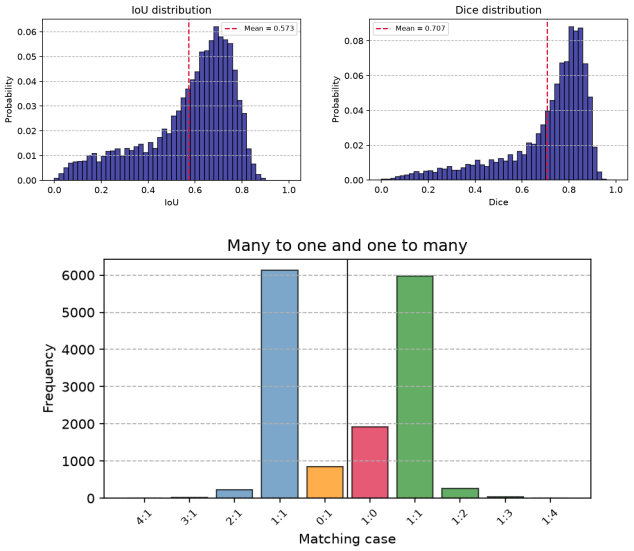


Figure 3: Out-of-fold IoU and Dice distributions over GT/prediction pairs with IoU > 0, and the one-to-many / many-to-one counts (left: number of GT linked to each prediction; right: number of predictions linked to each GT).

Table 2: Variants (fold 0, 142 images, tuned PQ) and post-processing rules (5-fold OOF, 707 images).

Variant (fold 0)	PQ
U-Net ResNet-34, 40 epochs (baseline)	**0.416**
10 / 20 epochs	0.389 / 0.396
+ crop refiner at native resolution	0.413
+ Mask R-CNN instances not covered	0.417
Mask R-CNN alone (disjoint masks)	0.377
ResNet-50 encoder	0.414
Limb-darkening-corrected input	0.411
BCE + Lovász hinge	0.390
Average of ResNet-34 and ResNet-50	0.416
8 dihedral views (4 flips: 0.419)	0.419
2048 px, folds 0 / 3 (1536 px: 0.403 / 0.387)	0.404 / 0.388
Post-processing rule (OOF, 1024 px)	**PQ**
Submission setting (confidence ≥ 0.8)	0.409
Confidence ≥ 0.85	0.411
No closing, no confidence or limb filter	0.398
Size-dependent confidence threshold	0.411
Hysteresis (seed and grow thresholds)	0.409
Masks dilated / eroded by 1 px	0.381 / 0.399
70-degree limb cut-off	no change

3.3 Where the remaining PQ is lost

Near misses. Of the 3,174 OOF false negatives, 45% overlap no prediction, 18% overlap one at IoU < 0.3 and 37% at IoU 0.3–0.5. Among the near misses the prediction is smaller than the GT in 46% of cases and larger in 34%; only 17% are merges with another filament and 16% splits. Dilating or eroding every mask by one pixel costs 0.01–0.03 PQ, because filaments are only a few pixels wide: the extent errors are per instance, not a bias a rule could correct. *Small filaments.* Recall falls to 0.36 below 500 px. Annotators rarely agree on them: the share of other annotators marking the same filament (IoU > 0.5) is 0.28 below 300 px and about 0.60 above 1,000 px. *When a doubtful prediction pays.* A prediction matching k of n annotators raises PQ only if $k/n > \mathrm{PQ}/(2\,\mathrm{SQ}) \approx 0.31$. A filament marked by one annotator of three is break-even, which is why lowering the area or confidence thresholds never helped; 28% of our false positives match another annotator's filament on the same image.

3.4 Strengths, weaknesses and possible improvements

Strengths. One network and a small tree model, no external data, 4.5 s per image, and a submission reproduced byte for byte by the notebook. Out of fold it exceeds a single annotator on both boundary quality (SQ) and detection (RQ): trained on the consensus, it predicts the filaments most annotators agree on. The scorer optimises the metric directly and needs no hand-tuned thresholds. *Weaknesses.* Missed annotations are now the largest error (23% of annotations overlap no prediction), mostly small filaments and filaments marked by one annotator; extent errors of one or two pixels on thin filaments cost SQ; and the gain over the previous pipeline is small relative to the leaderboard's resolution. *Improvements.* Since most of the remaining loss is disagreement between annotators, further gains should come from modelling the annotators (for example a head per annotator, or predicting each annotator's style) rather than from a stronger segmenter; a second-stage scorer could also be given image context beyond the two image-level features.

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